

Ari Scheuer

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Education

B.S. Mechanical Engineering | Milwaukee School of Engineering | Graduating May 2028

Internship Experience

Advanced Automation Engineering Intern | Komatsu Mining | Jun 2026 – Present

Project Objective:

- Design, manufacture, and program a 1/5th-scale 7-DOF mockup robotic track-drill arm for forward kinematic testing, production sensor qualification, and training.

Project Results:

- Built and programmed the mockup arm using 3D-printed components driven by linear actuators and servo motors, controlled via LabVIEW on a cRIO RT/FPGA target. The control panel supports manual operation and programmed motion sequences, using rotary and incremental encoder feedback to calculate real-time drill bit position in 3D space.

Embedded Systems Design Associate | CT Cubed Inc | May 2025 – Aug 2025

Project Objective:

- Developed a portable hardware-in-the-loop simulation testbed demonstrating cyberattacks on autonomous drone systems. Engineered a ruggedized enclosure with modular 3D-printed fixtures to streamline replication and reduce cost.

Project Results:

- Integrated UAV hardware, BLDC motors, LiDAR, CV cameras, and environmental sensors into a unified HIL platform.
- Developed a C++ backend for real-time hardware and system control and a Python GUI for live telemetry, enabling observation of system response under simulated attacks.

Leadership & Project Experience

President | MSOE High Powered Rocket Club | May 2026 – Present

Two-Stage Rocket Project Manager | MSOE High Powered Rocket Club | Sep 2025 – Mar 2026

Project Objective:

- Led a team of 14 to design, develop, and construct a two-stage high-powered rocket for the 2026 Argonia Cup Competition.
- Validated system designs on a half-scale testbed rocket; ensured successful staging, ignition, and safe dual-deployment recovery.

Project Results:

- Led system integration across structural, avionics, and payload subteams to ensure full functionality.
- Performed aerodynamic and physics simulations in MATLAB and OpenRocket to validate the rocket's stability and flight performance.
- Validated staging, ignition, and recovery on a half-scale testbed; launched to 3,003 ft AGL and recovered safely.
- Optimized motor selection by modeling the competition scoring equation in MATLAB and Excel, then validated candidate motor pairs in OpenRocket to identify the highest-scoring configuration.
- Staged, ignited, and recovered from an altitude of 12,000 ft AGL, placing 6th out of 21 teams at the 2026 Argonia Rocket Cup.

Autonomous Self-Driving 1972 Citicar | Personal Project | Mar 2025 – Present

Project Objective:

- Restored the chassis, body, and mechanical systems while re-engineering the powertrain and vehicle electronics.
- Developed an autonomy-ready control architecture, validated through closed-loop simulation before road testing.

Project Results (ongoing):

- Refurbished the original 3.5 hp 48V motor and integrated a Curtis 1205M motor controller.
- Designed a 48V battery from two Tesla Model S modules, delivering a 50-mile range.
- Reverse-engineered the controller's proprietary RS232 telemetry protocol and developed a microcontroller gateway that translated live data into CANopen frames on the vehicle CAN bus.
- Built a kinematic vehicle model in Simulink using the Vehicle Dynamics Blockset, parameterizing body, tire, brake, and motor subsystems with Citicar-specific data.
- Integrated the autonomy sensor stack with the Simulink model in closed-loop simulation and tuned a Kalman filter to quantify 3D state-estimation accuracy before on-vehicle testing.

Technical Skills

Hardware Skills: SolidWorks, Fusion360, OnShape, CNC programming, 3D printing, PCB Design, Automated systems

Software Skills: MATLAB, Simulink, LabVIEW, Python, C++, Mechatronics, Embedded systems, MS Office, Excel, HIL, CANopen, Git